Earth and Planetary Science Letters, 101 (1990) 272–280
Elsevier Science Publishers B.V., Amsterdam

[XLeP]

Paleomagnetic evidence for rotation during Stephano-Permian extension in southern Massif Central (France)

J.P. Cogné [a], J.P. Brun [a] and J. Van Den Driessche [b]

[a] *C.A.E.S.S., Institut de Géologie, Université de Rennes 1, F-35042 Rennes-Cedex, France*
[b] *Laboratoire de Tectonique, Université de Paris VII, F-75230 Paris-Cedex 05, France*

Received February 2, 1990; revised version accepted June 28, 1990

ABSTRACT

We present results of a preliminary paleomagnetic study devised as a test of a tectonic model proposed by Van Den Driessche and Brun [1] to interpret the main structural features of the Montagne Noire (South Massif Central, France). This model suggests that crustal extension during Stephanian to Permian times is responsible for a major flat-lying detachment. The inferred geometrical relationships between fault azimuth and extension direction is believed to induce a counter-clockwise rotation of the Paleozoic cover during its northward motion. In order to test this model, we have sampled Autunian (Lower Permian) grey and red sandstone beds within two basins opened during the extension, the Lodève and the St-Affrique basins. The paleomagnetic study of six sites (three in each basin) using thermal demagnetization procedures, reveals the presence of 3 magnetization components. A first one (A), which is unblocked below 400 °C, is interpreted as a recent overprint due to weathering of the sandstones. The B component is unblocked between 400 ° and 575 ° C. This B component is assumed to be primary, although the gentle dip of the beds does not allow a fold test. It fits the Lower Permian field direction in the Lodève basin (*in situ*: $D_m = 194°$, $I_m = 6°$, $k = 50$, tilt corrected: $D_m = 194°$, $I_m = -10°$, $k = 53$). In contrast, it shows a systematic difference in declination in the St-Affrique basin (*in situ*: $D_m = 173°$, $I_m = 2.5°$, $k = 101$, tilt corrected: $D_m = 172°$, $I_m = 6°$, $k = 37$). This is interpreted as a post-Autunian counter-clockwise rotation of at least 20 ° of the St-Affrique basin, thus supporting one of the main implications of the tectonic model. Finally, red sediments from two sites of the Lodève basin yield a third component (C) with unblocking temperatures up to 670 ° C and carried by hematite. Its direction is also consistent with the Permian field direction, showing that hematite crystallisation occurred early in the diagenesis history of the sediments.

1. Introduction

After a long period of crustal thickening from Silurian to Visean times [2,3], the French Variscan Belt has undergone extension in a comparable way as observed in the Basin and Range Province [1,4]. Associated sedimentary basins of Stephanian to Permian age are locally controlled by single flat-lying detachments (e.g., see Mattauer et al. [5]). In the Montagne Noire region (Fig. 1), the extensional system which separates an upper plate from a lower plate consists of an E–W-striking, north-dipping detachment (Espinouse) and of two oblique transfer faults (Plaisance and Cévennes) which strike roughly parallel to the NE direction of extension [1]. The lower ductile crust of the lower plate is exhumed, forming a gneiss dome: the Montagne Noire Axial Zone. The St-Affrique and Lodève basins deposited on the upper plate. They are separated by the imbricate units of the Monts de Lacaune. The NE-striking and steeply dipping faults of these units are classically interpreted as thrusts [6,7]. However, they appear anomalous with respect to the general E–W strike and shallow dip of thrusts observed in the Southern Massif Central. This is interpreted by Van Den Driessche and Brun [1] as the result of a counter-clockwise rotation of the upper plate (Monts de Lacaune) during extension, due to the non-orthogonality of the main detachment (Espinouse) with respect to the bulk extension direction deduced from the azimuth of transfer faults (Plaisance and Cévennes; Fig. 1). This counter-clockwise rotation has induced a progressive steepening of the Monts de Lacaune thrusts and their reorientation parallel to the extension direction.

The above tectonic model is here compared with results of paleomagnetic investigations because paleomagnetism is a powerful means of

0012-821X/90/$03.50 © 1990 – Elsevier Science Publishers B.V.

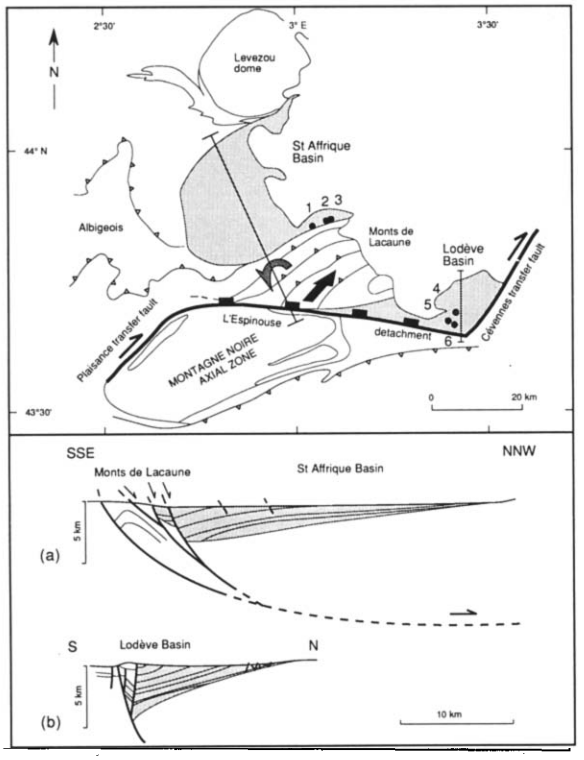


Fig. 1. Top: Schematic tectonic map of the studied area, with location of sampling sites, labelled *1* to *6*, in the St-Affrique and Lodève basins (grey areas). The large black arrow indicates the direction of extension on the Espinouse normal fault. The curved grey arrow indicates the suspected rotation of the Monts de Lacaune basement rocks. Bottom: schematic cross-sections, located on the map, in the St-Affrique basin and the Monts de Lacaune (a), and in the Lodève basin (b), showing the onlap structure of the basins (map and cross-sections after Van Den Driessche and Brun [1]).

checking relative rotations of crustal blocks. However, a serious problem was raised in finding convenient formations for undertaking this test. Indeed, the Paleozoic basement rocks of the Monts de Lacaune area, where the largest rotations are expected, are made of slates, sandstones and limestones strongly folded and thrusted during the main Hercynian phases. In contrast, the low-strain domains constituted by the Permian basins of St-Affrique and Lodève appeared more convenient for paleomagnetic analysis.

Because these basins were opening during the late Paleozoic extension, we could make the following predictions: (1) being located at the corner between the main detachment and the Cévennes transfer fault, the Lodève basin has suffered no, or minor rotation; (2) in contrast, the basin of St-Affrique should rotate during Permian sedimentation because its basement was undergoing rotation. A preliminary sampling of the lowermost part of both basins was thus carried out. Our results will be discussed together with other paleomagnetic studies recently done within the Lodève basin [8,9].

2. Paleomagnetic analysis

Forty-eight cores were drilled and oriented at six sites within Autunian (Lower Permian 288–272 Ma) beds of St-Affrique (sites p1, p2 and p3) and Lodève (sites p4, p5 and p6) basins (Mean coordinates: 43.8° N, 3.3° E; Fig. 1). These beds are grey shales and sandstones in the lower part, and typical red beds in the upper part. However, field studies have shown that red zones cut across bedding planes (e.g., see Clement [10]). Therefore, precipitation of hematite, which gives the red color to the beds, is a secondary event with respect to sedimentation. Two sites (p4 and p5) have been

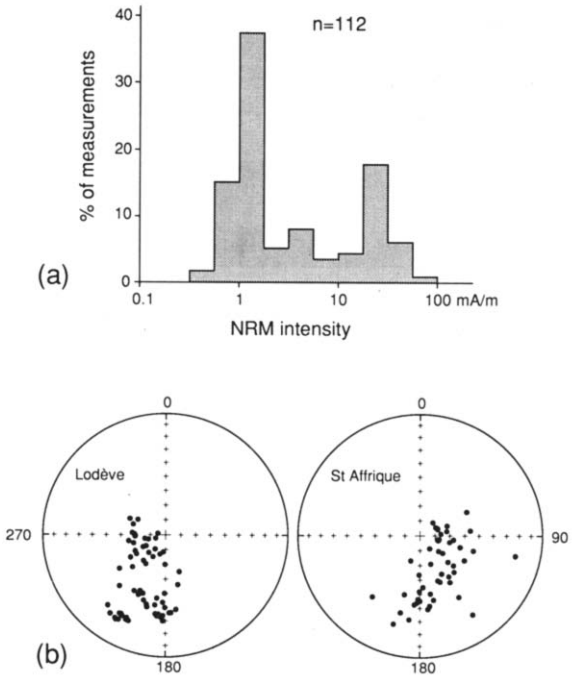


Fig. 2. NRM data. (a) Intensity frequency plot. (b) NRM directions measured in specimens from Lodève (left) and St-Affrique (right) basins. Equal-area projections of the lower hemisphere.

TABLE 1

Mean site structural and paleomagnetic data [a]

Site	Bedding	B component					C component				
	S/D	n/no	IS	TC	k	α_{95}	n/no	IS	TC	k	α_{95}
p1	278/14	8/8	172/3.5	171/17	43	8	–	–	–	–	–
p2	168/10	8/8	173/10	174/9	123	5	–	–	–	–	–
p3	160/16	7/8	173/−6	171/−9	159	5	–	–	–	–	–
p4	77/21	8/8	199/−1	201/−19	54	7.5	8/8	201/−10.5	206/−27	204	4
p5	84/19	8/8	191/18	190/0	45	8	8/8	186/19	185/1	61	7
p6	84/14	6/8	192/1	193/−12	70	8	–	–	–	–	–
Means											
St-Affrique		3	173/2.5		101	12					
				172/6	37	21					
Lodève:											
(this study)		3	194/6		50	17	2	195/4			
				194/−10	53	17			195/−13		
(this study		10	192/14		82	5.5					
+[8,9])				191/0	53	6.5					

[a] Bedding plane attitude is given as strike (S)/dip (D); Mean site directions of B and C: n/no: number of entries in the statistics/number of studied specimens; IS and TC: declination/inclination of *in-situ* and tilt-corrected mean site directions of magnetization, respectively; k, α_{95}: Fisher [16] statistics parameters. Two formation mean directions of B component are given for Lodève: the first is computed from our data ($n = 3$), the second includes sites FUPA and FMPA of Courtillot et al. [8], and sites 5, 12, 14, 16, 17 of Merabet and Guillaume [9] ($n = 10$).

sampled in the red beds; the others are in grey sandstones.

These Permian formations have not been folded. Nevertheless, the development of the basin during sedimentation induced a progressive tilt of the beds as sketched on cross-sections of Fig. 1. Bedding orientations at sampling sites are given in Table 1.

All measurements of remanent magnetization have been done with either a Schonstedt DSM-1 spinner magnetometer, or a LETI cryogenic magnetometer, depending on magnetization intensities.

2.1. *Natural remanent magnetization (NRM)*

NRM has been measured on 112 specimens from the 48 cores. Results are illustrated in Fig. 2. Two groups of NRM intensities appear on a frequency histogram (Fig. 2a). One group averaging 30–50 mA/m is from red sandstones (sites p4 and p5). Samples from the other sites show weak intensities of the order 1 mA/m. NRM directions (Fig. 2b) are partitioned in two groups which are not related to the nature of the beds, but instead to the location of sampling sites. Indeed, if both populations show an elongate distribution from shallow southerly directions to highly inclined ones, NRM directions from the Lodève basin are mainly situated in the SW quadrant of the projection, whereas directions from the St-Affrique basin cluster in the SE quadrant. This suggests (1) overprinting of a primary Permian paleomagnetic component (roughly south and horizontal) by a secondary one in the present-day field (north and steep) and (2) a systematic counter-clockwise deviation of the declination within the St-Affrique samples with respect to the declination in samples from Lodève.

2.2. *Demagnetizations*

Standard stepwise thermal demagnetization procedures have been applied to eight specimens per site using a Schonstedt TSD-1 furnace. Typical demagnetization curves are shown in Fig. 3. They reveal the multicomponent nature of magnetization in these beds. Apart from a viscous component which is removed below 100 °C, a first component (A) is demagnetized between 100 and 300–400 °C. It has northerly declinations with steep inclinations. As shown by Fig. 3, this component can be more or less well isolated. The second component (B) has typical unblocking

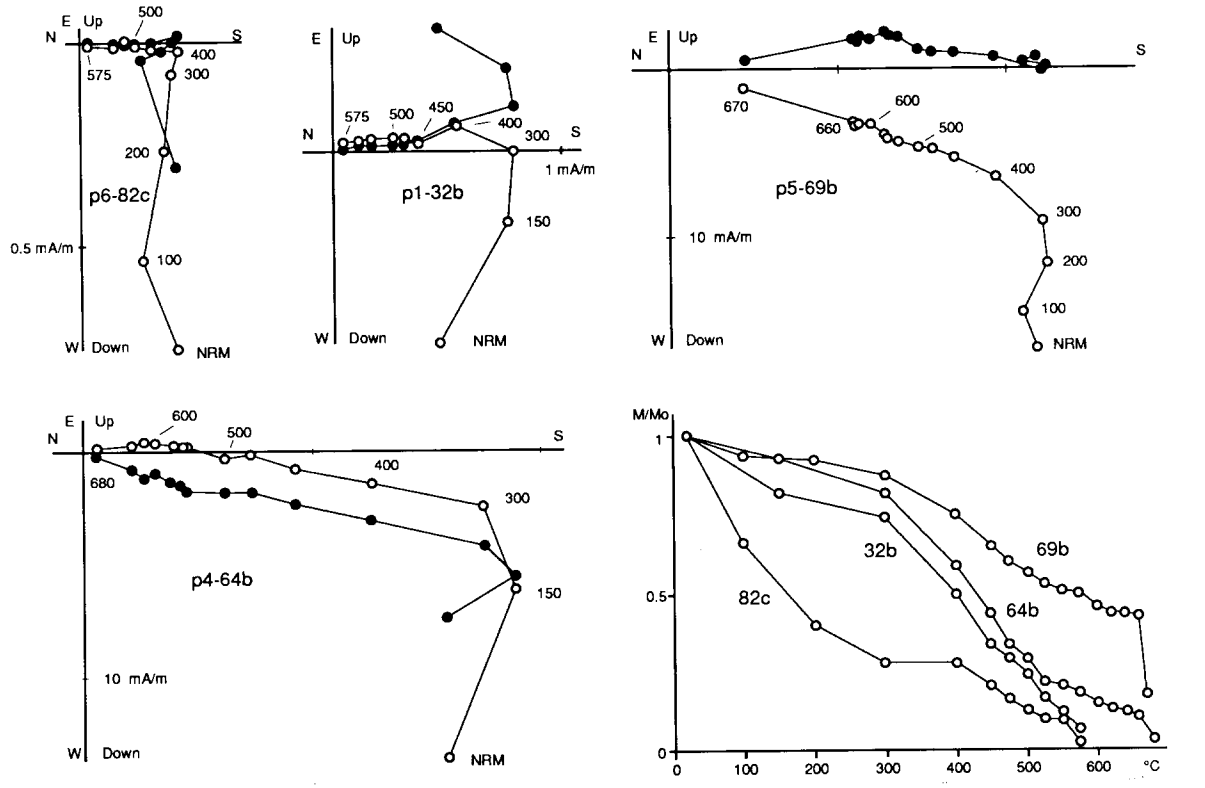


Fig. 3. Typical thermal demagnetization curves in orthogonal projections [15], and normalized intensity decay curves for four samples from Lodève and St-Affrique Autunian sandstones. Closed (open) symbols: projection onto the horizontal (vertical) plane. Temperatures in °C.

temperatures ranging from 400 to ~570°C. It has south declination and very shallow positive to negative inclination.

Samples from sites p1, p2, p3 and p6 in grey sandstones exhibit components A and B only. In contrast, red beds from sites p4 and p5 show a third component (C) with high unblocking temperatures, ranging from 600 to 680°C (Fig. 3, specimens p5-69b and p4-64b). Although directions of B and C differ only by a few degrees, it is clear that the demagnetization path is not linear in these samples, but shows a change in direction above 600°C. This could eventually be interpreted as due to a bad separation of the low-temperature (100–300°C) A component and the high-temperature (600–680°C) C component. If so, the B component in these samples could be an intermediate direction between A and C, without any paleomagnetic significance. However, the good linearity of the demagnetization path between 400 and 600°C, and the stability in intensity between 600 and 660°C followed by a sharp breakdown at 670–680°C (Fig. 3) lead us to assume that red sediments actually bear two high-temperature magnetization components carried by two different magnetic minerals, one with unblocking temperatures below 600°C, the other with unblocking temperatures up to 680°C. As shown below, this is consistent with our interpretation of IRM acquisition experiments.

2.3. Isothermal remanent magnetization (IRM)

IRM acquisition experiments in fields up to 1.2 T were performed on one specimen from each site using a Brucker BE-10 electromagnet. IRM acquisition curves and coercivity spectra are shown in Fig. 4. Two types of results were obtained. A first type is represented by specimens from sites p3 and p6 (Fig. 4, bottom), where 60 to 70% of IRM at 1.2 T is acquired below 0.1 T. Although this low-coercivity phase is dominant, there is a continuing acquisition by a high coercivity one up

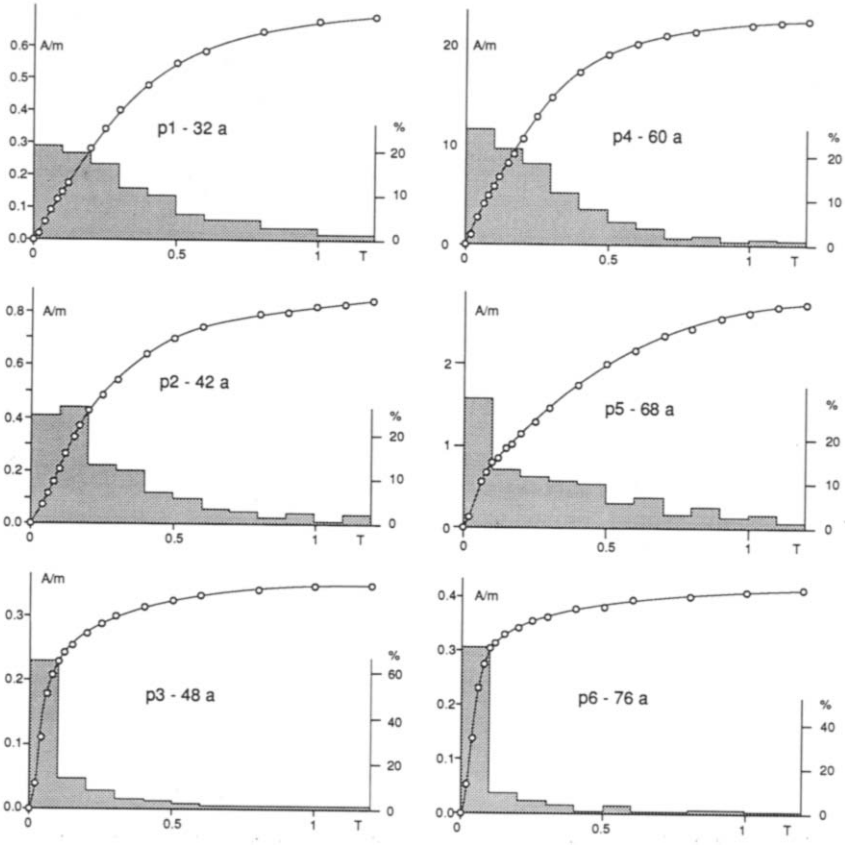


Fig. 4. IRM acquisition curves and coercivity spectra (in grey) for one sample of sites *p1* to *p6*. Field in Tesla (T), left scale: IRM intensity in A/m, right scale % of IRM at 1.2 T.

to fields of 1.2 T. In the second type, from the other sites, the coercivity range is larger, with significant IRM acquisition up to 1 T fields. It may be further noted that specimens p4-60a and p5-68a show a mixed type. This suggests a probable coexistence of both low- and high-coercivity phases in all sites, thus supporting the previous assumption of at least two magnetic phases.

3. Analysis of paleomagnetic directions

3.1. A components

The low-temperature components are plotted in *in-situ* coordinates in Fig. 5. They cluster around the present-day dipole field axis (open star in Fig. 5) and are therefore thought to be a recent remagnetization, probably of weathering origin. The low unblocking temperatures suggest goethite as the carrier of this component, which is consistent with the high coercivities observed in all sites.

3.2. B component

The B component is defined in all sites by an unblocking temperature ranging from 400 to 550–

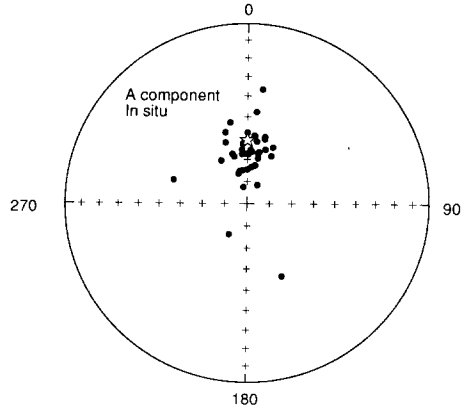


Fig. 5. *In-situ* directions of low-temperature (A) components. Star: present-day dipole field axis. Equal-area projection of the lower hemisphere.

600 °C. This suggests that it is carried by magnetite, which is consistent with the low IRM coercivity magnetic mineral recognized in two sites and suspected in the others. Site mean directions of this component are given in Table 1 and illustrated in Fig. 6. This Figure shows projections in a vertical E–W plane, where site mean directions of the B component are compared to the path of directions obtained from Lower to Upper Permian apparent polar wander path of Piper [12] for Stable Europe (little open dots) and from Lower and Upper Permian VGP of Westphal et al. [13] (asterisks).

Our data from the Lodève basin (sites p4, p5 and p6) are clearly superimposed on these Permian reference curves. It is therefore assumed that the B component is of Permian age. In contrast, directions of the B component from the St-Affrique basin show a significant departure from the reference curve, mainly due to a counter-clockwise deviation in declination of about 20 ° in average, whereas inclinations are similar to those from Lodève. According to similarities in rock type and age, and in magnetization behaviour during demagnetization, it is assumed that the B component from St-Affrique is also of Permian age.

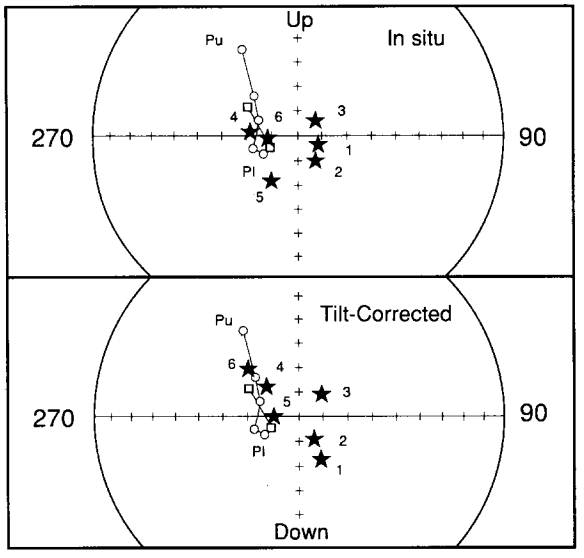


Fig. 6. Intermediate temperature B components. Vertical E/W equal area projections of the southern hemisphere. Stars *1* to *6*: mean site directions of B components from sites p1 to p6. Open circles: field directions deduced from Piper [12] APWP for stable Europe between Lower (*Pl*) and Upper (*Pu*) Permian. Open square: field directions from Westphal et al. [13] Lower and Upper Permian VGPs.

This hypothesis could not be verified using the classical fold test [11]. Including paleomagnetic data of Courtillot et al. [8] (two sites) and Merabet and Guillaume [9] (five sites) obtained in Autunian grey sediments from Lodève, a fold test may be performed on 10 site mean directions (Table 1). The k parameter decreases from 82 *in situ* to 53 after tilt correction, which is not significant at the 95% probability level. This inconclusive fold test may have two main causes. First, all the bedding planes are weakly dipping towards the south (Table 1 and [8,9]). Therefore, between site dispersion cannot be strongly modified by unfolding. A second cause may be found in the fact that the tilting of bedding planes is not induced by a later folding event but occurred during sedimentation, and thus more or less during magnetization acquisition processes. In such a situation, magnetization may be acquired either by already tilted beds or in the course of extension-induced tilting processes. This interpretation is supported by the following observations: (1) the k parameter tends to decrease upon tilt correction (Table 1); (2) the tilt correction seems to induce a scatter in inclination between Lodève and St-Affrique data; (3) the between-site dispersion of both Lodève and St-Affrique data is greater in inclination than in declination (Fig. 6) for *in-situ* as well as for tilt-corrected data; (4) the *in-situ* directions from Lodève are closer to the Lower Permian part of the reference curve than the tilt-corrected ones. This leads us to the conclusion that tilt-corrected directions of the B component do not necessarily conform the actual Lower Permian field direction in the Lodève basin. This contrasts with the conclusions of Courtillot et al. [8] (but based on only two sites), and of Merabet and Guillaume [9] who used tilt-corrected directions without arguing this choice. However, we note that the difference in declination between Lodève and St-Affrique data is observed in either *in-situ* or tilt-corrected coordinates. This feature is thus not related to possible problems of tilt correction.

Finally, the difference in declination cannot arise from a later magnetization acquisition age in the St-Affrique basin, since the evolution of the Permian field direction affects inclination only (Fig. 6). Hence, it is reasonable to conclude that

the B component from both basins is an early magnetization acquired at Early Permian times. Therefore, the counter-clockwise deviation of magnetization may reflect a counter-clockwise rotation of the St-Affrique basin with respect to the Lodève basin, as predicted by the tectonic model [1].

3.3. C component

The C component has been isolated in red beds of sites p4 and p5 only. The high unblocking temperatures where this component is defined show that it is carried by hematite. This is consistent with the red colour of the beds and with the high-coercivity carriers observed in IRM experiments. Site mean directions of the C component are given in Table 1 and illustrated in Fig. 7 where they are compared to site mean directions of the B component for both p4 and p5 sites. It is obvious in each site that both directions are very close to each other. This is particularly clear for site p5: the test of Watson [14] shows that populations of B and C are identical, although these components may be separated in demagnetization diagrams. This is not observed at site p4 where the two populations are distinct. However, C and B directions remain very close to each other, and in both sites, C directions are consistent with an Early to Middle Permian age, in either *in-situ* or tilt-corrected coordinates. We finally note that the C component has not been identified by Merabet and Guillaume [9] in their red Autunian samples. However, a careful examination of the orthogonal projection of their Fig. 4 shows that the demagnetization path converges towards the origin only after 600 °C, whereas the 400–600 °C range shows a slightly different direction. This lends indirect support to our determination of this component.

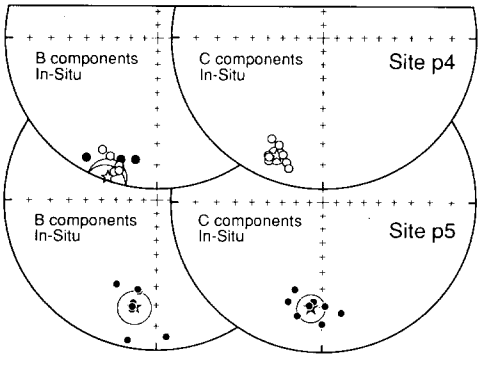


Fig. 7. *In-situ* directions of B (left) and C (right) components in sites *p4* (up) and *p5* (down) with their within-site means (stars) and circles of 95% confidence. Open (closed) symbols: negative (positive) inclination. Stereographic projections.

The unconformity between the reddening front and bedding surfaces suggests that hematite crystallisation is not synchronous with sedimentation [10]. This is further outlined by our paleomagnetic data which show that, although diachronous, the formation of hematite occurred early in the diagenetic history of the sediment, not after Late Permian times. Furthermore, the angular relationship between B and C at site p4 (B subhorizontal, C with the same declination and slightly higher negative inclinations) is consistent with the evolution of field directions during the Permian. This support the secondary, albeit early, origin of C component.

Because of the small number of sites, the lack of fold test, and the lack of precise timing of beds tilting, the above discussion necessarily remains general. Only a wider sampling should allow to precise these relationships between B and C components. This, however, does not question our demonstration of a relative rotation of the St-Affrique basin with respect to the Lodève basin.

4. Conclusions

We have presented the results of a paleomagnetic study devised as a test of Permian rotations predicted by a kinematic model of extension in the Montagne Noire area [1]. For this, we have sampled Lower Permian (Autunian) sandstone beds in two basins, Lodève and St-Affrique, opened during the late Paleozoic extension. The paleomagnetic analysis through thermal demagnetization procedures allows to draw the following conclusions.

A low-temperature component (A) clusters around the present-day dipole field axis, and is interpreted as a remagnetization due to weathering of the beds. After removal of this component, the characteristic remanent magnetization component (B) is isolated between 400 and 570 °C. This component, which is probably carried by magnetite, is interpreted as being the primary Lower Permian magnetization in both basins, on the basis of a comparison with Permian APWP for Sta-

ble Europe. However, whereas the B component from the Lodève basin fits the Permian field direction, data from St-Affrique show a significant difference in declination averaging 20°. This provides evidence for a counter-clockwise rotation of the St-Affrique basin with respect to the Lodève basin (Fig. 8). Taking into account that extension started during the Stephanian, and that the paleomagnetic analysis has been done on Autunian sediments, the declination difference of 20° gives an estimate of the minimum amount of rotation associated with extension. If compared with the mean E–W trend of thrusts in Southern Massif Central, the peculiar NE–SW orientation of the Monts de Lacaune thrusts suggests a total rotation of up to 45° (that is, from E–W to their present direction). During this rotation, thrust planes have been steepened and reactivated as oblique normal fault with a sinistral component of strike slip, indicating that thrust slices have moved against each other like dominoes (Fig. 8).

Apart from the above conclusion which is the main result of this study, our analysis allows to address two additional points: (1) The third component (C) carried by hematite in red beds from the Lodève basin also conforms the Permian field direction for Stable Europe. This shows that reddening processes (i.e., hematite crystallization) took place early in the sediment history. However, the C component is interpreted as secondary with respect to the earlier B component. (2) The unsuccessful fold test for the Lodève Autunian paleomagnetic directions, supposed to be related to problems of relative timing of magnetization acquisition and bed-tilting processes, indicates that tilt-corrected paleomagnetic directions may not accurately define the Lower Permian field direction. This point is critical, since the reference APWP for Stable Europe largely relies on data from this basin. It is therefore concluded that the uncertainties on the poles deduced from these data may be larger than the conventional cones of confidence.

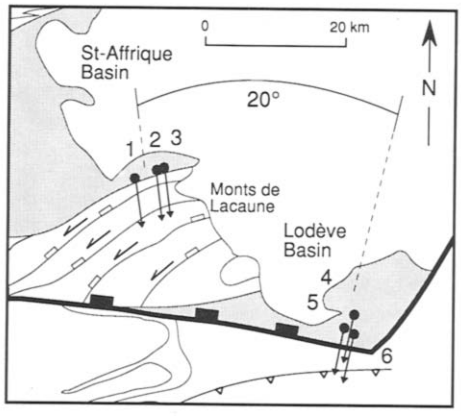


Fig. 8. Schematic map of the studied area with paleomagnetic declination (arrows) of *in-situ* B component determined in each of sites *p1* to *p6*.

Acknowledgements

Laboratory measurements and experiments have been done by F. Calza. This is a contribution of C.A.E.S.S. (UPR CNRS 4661).

References

1 J. Van Den Driessche and J.P. Brun, Un modèle cinématique de l'extension paléozoique supérieur dans le Sud du Massif Central, C.R. Acad. Sci. Paris 309, 1607–1613, 1989.
2 J.P. Burg and P. Matte, A cross-section through the French Massif Central and the scope of its Variscan evolution, Z. Dtsch. Geol. Ges. 129, 429–460, 1978.
3 P. Matte, Tectonics and plate tectonics model for the Variscan belt of Europe, Tectonophysics 126, 329–374, 1986.
4 G. Ménard and P. Molnar, Collapse of a Hercynian Tibetan Plateau into a Late Paleozoic European Basin and Range province, Nature 334, 235–237, 1988.
5 M. Mattauer, M. Brunel and P. Matte, Failles normales ductiles et grands chevauchements. Une nouvelle analogie entre l'Himalaya et la chaîne Hercynienne du Massif Central Français, C.R. Acad. Sci. Paris 306, 671–676, 1988.
6 P. Matte, The geotraverses across the Ibero-Armorican Variscan Arc of Western Europe, Geodyn. Ser. AGU 10, 53–81, 1983.
7 J.P. Burg, P. Balé, J.P. Brun and J. Girardeau, Stretching lineation and transport direction in the Ibero-Armorican Arc during the Siluro-Devonian collision, Geodin. Acta 1, 71–87, 1987.
8 V. Courtillot, P. Chambon, J.P. Brun, P. Rochette and P. Matte, A magnetotectonic study of the Hercynian Montagne Noire (France), Tectonics 5, 733–751, 1986.
9 N. Merabet and A. Guillaume, Paleomagnetism of the Permian rocks of Lodève (Hérault, France), Tectonophysics 145, 21–29, 1988.
10 J.Y. Clement, Minéralogie, pétrologie et géochimie du Permien de Lodève (Hérault, France), E.N.S.M.P. Mém. Sci. Terre 2, 137 pp., 1986.
11 M.W. McElhinny, Statistical significance of the fold test in paleomagnetism, Geophys. J. R. Astron. Soc. 8, 338–340, 1964.
12 J.D.A. Piper, Palaeomagnetism and the continental crust, 434 pp., Wiley, New York, NY, 1987.

13 M. Westphal, M.L. Bazhenov, J.P. Lauer, D.M. Pechersky and J.C. Sibuet, Palaomagnetic implications on the evolution of the Tethys belt from the Atlantic ocean to the Pamir since the Triassic, Tectonophysics 123, 37–82, 1986.
14 G.S. Watson, Analysis of dispersion on a sphere, Mon. Not. R. Astron. Soc., Geophys. Suppl. 7, 153–159, 1956.
15 J.D.A. Zijderveld, Demagnetization of rocks: analysis of results, in: Methods in Paleomagnetism, D.W. Collision, K.M. Creer and S.K. Runcorn, eds., pp. 254–286, Wiley, New York, NY, 1967.
16 R.A. Fisher, Dispersion on a sphere, Proc. R. Soc. London Ser. A 217, 295–305, 1953.